

Probability & Random Processes EE-2011

Department of Electrical Engineering

NATIONAL UNIVERSITY OF COMPUTER & EMERGING
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Agenda

- Introduction of Course
- Course Outline & Grading Policy
- Review of Basic Concepts in Probability and Statistics
- Concepts of Random Experiment & Sample Space
- Concept of an Event, their Intersection and Union



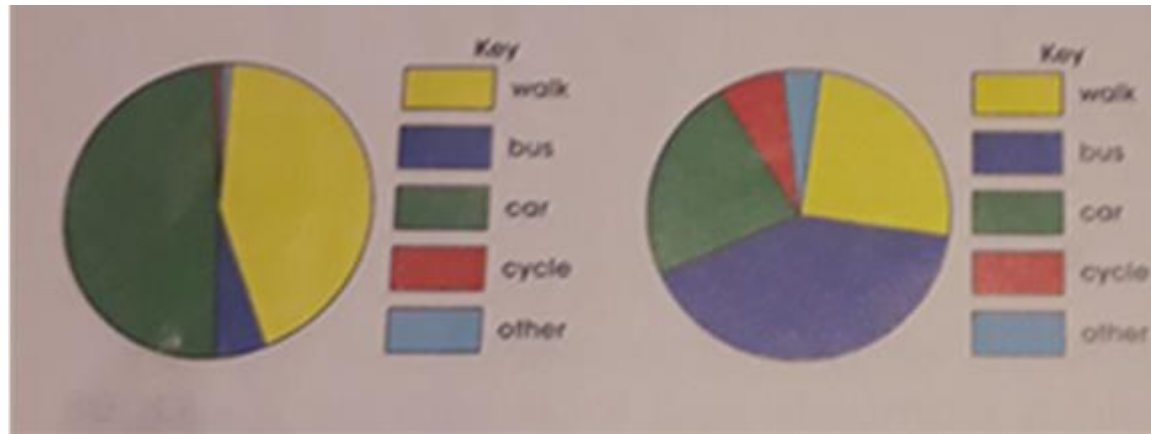
Living Lives and EE-2011

- How much certainty do you have in your lives?
- Do you anticipate future as you live?
- How many decisions did you make today?
- Did you ever regret any of your decision?
- Why did you?
- How could you improve the quality of your decisions?
- Will background information about the decision matter?
- How would you organize background information?
- What can you say about the certainty in information?
- What can you say about the variation in Information?
- How could you quantify certainty and variation in information?



Let's investigate

The council are planning transport for learners to and from school. This is what they found out about how 9 year old and 13 year old learners get to school.



Describe the difference between how 9 year olds and 13 year olds go to school. What could be the explanations for these differences?

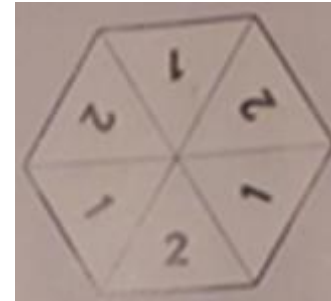
Vocabulary

statistics: the collection,
organization,
presentation,
interpretation and
analysis of data.



Let's investigate

What is the chance of spinning a '3' on the hexagonal spinner?



What is the chance of spinning a '2' on the hexagonal spinner?

What is the chance of spinning a '6' on the pentagonal spinner?



Some Vocabulary

- Chance?
 - likelihood that a particular outcome will occur
- Probability?
 - chance that a particular outcome will occur considering the total possible outcomes
- Likelihood?
 - The chance that a particular outcome will occur
- Impossible?
 - An event that has no chance of occurring; has probability of zero.
- Certain?
 - An event that will definitely occur; has probability of one.



Some Vocabulary

- Likely?
 - An event that will probably occur; has probability between $1/2$ and 1 .
- Unlikely?
 - An event that will probably not occur; has probability between 0 and $1/2$.
- Evens?
 - An event that has same chance of occurring and not occurring; has probability of $1/2$.
- Equally Likely?
 - When the chance of differ outcomes is the same, e.g., rolling a dice has probability of $1/6$ for all outcomes



Statistics

- It's about the:
 - Collection
 - Organization
 - Presentation
 - Interpretation, and
 - Analysis of Information



Introduction

- Japanese were able to succeed -> use of *statistical methods* and statistical thinking among management personnel.
- Statistical methods are used in:
 - Manufacturing
 - Development of food products
 - Computer software
 - Energy sources
 - Pharmaceuticals
 - and many other areas
- Every where there is gathering of information or **scientific data**.
- Difference between collection of scientific information and **inferential statistics**.
- Scientific Judgments in the face of **uncertainty and variation**-> to improve **quality** of the process.



Introduction

- Is product density in the process were always the same and were always on target?
- Is there no patient-to-patient variability inherent in the response to the drug?
- Statisticians make use of fundamental laws of probability and statistical inference to draw conclusions about scientific systems.



Gathering of Information

- Happening for thousands of years? Why?
 - How to organize it?
 - Certainty about information?
 - Variation in Information?
 - What do you do with the information?



Samples and Observation

- Information is gathered in the form of **samples**, or collections of **observations**.
- Samples are collected from **populations**, which are collections of all individuals or individual items of a particular type.
- At times a population signifies a scientific system.

Example:

- Sampling process may involve collecting information on 50 computer boards sampled randomly from the process.
- What is population here?
- Experimental Design vs **Observational study**



Example:

Data associated with a study conducted at the Virginia Polytechnic Institute and State University on the development of a relationship between the roots of trees and the action of a fungus. Minerals are transferred from the fungus to the trees and sugars from the trees to the fungus. Two samples of 10 northern red oak seedlings were planted in a greenhouse, one containing seedlings treated with nitrogen and the other containing seedlings with no nitrogen. All other environmental conditions were held constant. All seedlings contained the fungus *Pisolithus tinctorus*. The stem weights in grams were recorded after the end of 140 days.

No Nitrogen	Nitrogen
0.32	0.26
0.53	0.43
0.28	0.47
0.37	0.49
0.47	0.52
0.43	0.75
0.36	0.79
0.42	0.86
0.38	0.62
0.43	0.46

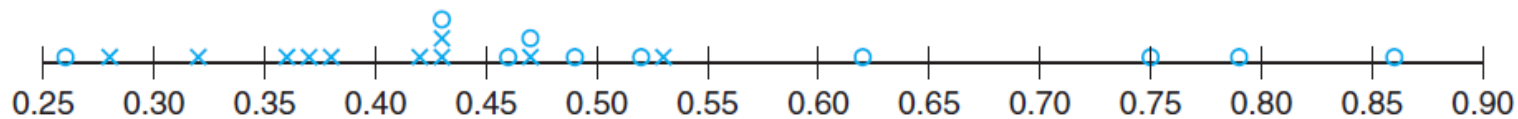


Figure 1.1: A dot plot of stem weight data.

Conclusions of the example: (p value)

- On average, the use of nitrogen increases the stem weight
 - Four nitrogen observations are considerably larger than any of the no-nitrogen observations.
 - Most of the no-nitrogen observations appear to be below the center of the data.
 - Appearance of the data set would seem to indicate that nitrogen is effective.
 - But how can this be quantified?
 - How can all of the apparent visual evidence be summarized in some sense?
-
- It may be summarized in a probability statement or P -value.
 - Suppose that this probability is small, say 0.03.



Simple Random Sampling

- **Simple random sampling** implies that any particular sample of a specified *sample size* has the same chance of being selected as any other sample of the same size.
- The term **sample size** simply means the number of elements in the sample.

What is a biased sample?

Example:

- The sample involves the choice of, say, 1000 families, and a survey is to be conducted. Now, suppose it turns out that random sampling is not used. Rather, all or nearly all of the 1000 families chosen live in an urban setting. It is believed that political preferences in rural areas differ from those in urban areas. In other words, the sample drawn actually confined the population and thus the inferences need to be confined to the “limited population,” and in this case confining may be undesirable. If, indeed, the inferences need to be made about the state as a whole, the sample of size 1000 described here is often referred to as a **biased sample**.
- Simple random sampling is not always appropriate



Stratified Random Sampling

- Sampling units are not homogeneous and naturally divide themselves into nonoverlapping groups that are homogeneous. These groups are called *strata*.

Example:

Suppose a sample survey is conducted in order to gather preliminary opinions regarding a bond referendum that is being considered in a certain city.

The city is subdivided into several ethnic groups which represent natural strata.

In order not to disregard or overrepresent any group, separate random samples of families could be chosen from each group.



Why Assign Experimental Units Randomly?

- Possible negative impact of not randomly assigning experimental units.
- Example: Among the characteristics of the patients that produce variability in the results are age, gender, and weight. Sample of people that are predominately heavier than those in the treatment group. Perhaps heavier individuals have a tendency to have a higher blood pressure.
- This clearly biases the result



Sample Mean and Median

Definition 1.1: Suppose that the observations in a sample are $x_1, x_2, \dots, x_n$. The sample mean, denoted by $\bar{x}$, is

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

Definition 1.2: Given that the observations in a sample are $x_1, x_2, \dots, x_n$, arranged in increasing order of magnitude, the sample median is

$$\tilde{x} = \begin{cases} x_{(n+1)/2}, & \text{if } n \text{ is odd,} \\ \frac{1}{2}(x_{n/2} + x_{n/2+1}), & \text{if } n \text{ is even.} \end{cases}$$

Example: Suppose the data set is the following: 1.7, 2.2, 3.9, 3.11, and 14.7 What are **Sample Mean and Median?**

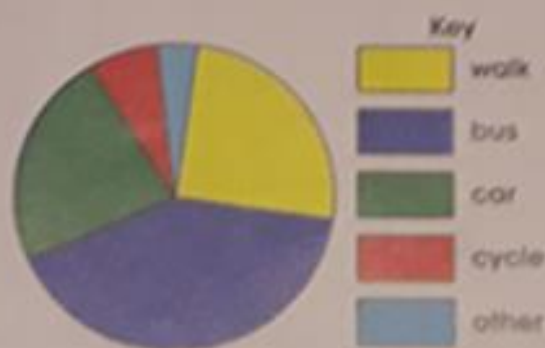


Statistics

Using statistics

Let's investigate

The council are planning transport for learners to and from school. This is what they found out about how 9 year old and 13 year old learners get to school.



Describe the difference between how 9 year olds and 13 year olds go to school. What could be the explanations for these differences?

Averages & Range

Average and range

Let's investigate

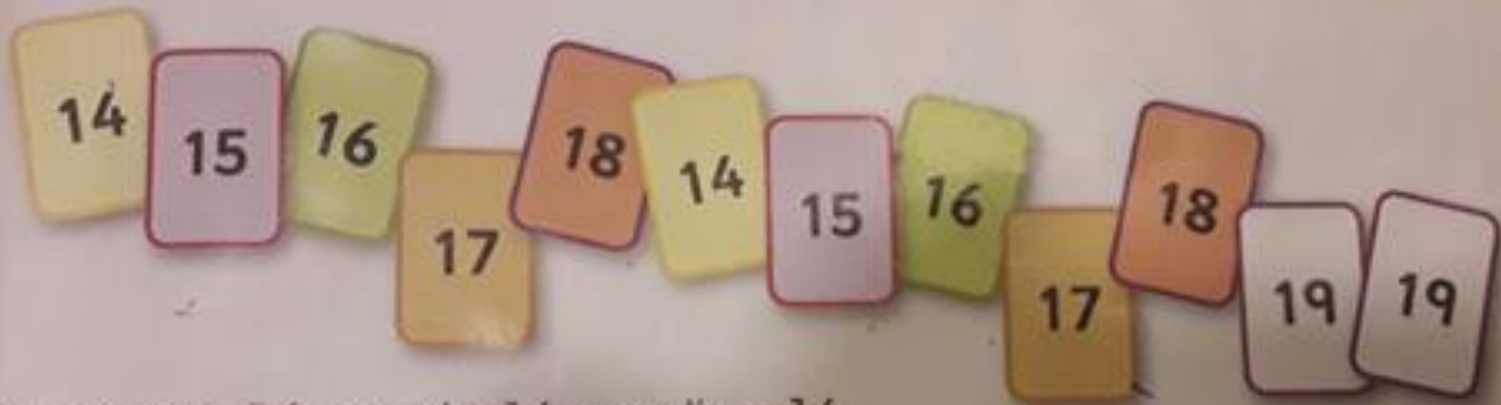


The mean average of the numbers on these cards is 5.

What number is on the fifth card?

Averages & Range

3 Use some of these cards to make sets of data to match these averages.



(a) mean 16, mode 16, median 16

(b) mean 17, mode 18, median 17.5 ✗

(c) mean 16, mode 15, median 15

Some Terminologies

- ***Experiment*** describes the process that generates a set of data
- **Some experiments** generate data that is subject to uncertainties
- Recording of information is ***Observation***
- ***Categorical data*** is classified according to some criteria
- Statistics is about “***Chance Outcome***” that occurs in a planned study or investigation



Some Terminologies

- There are three types of Statistical Studies
 - Designed Experiment
 - Observational Studies
 - Retrospective studies
- Designed Experiment
 - Design of experiments (DOE) deals with planning, conducting, analyzing, and interpreting controlled tests
 - to evaluate the factors
 - that control the value of a parameter.
- Observational Studies
 - An observational study draws inferences from a sample to a population



Statistical Studies

- where the independent variable is not under the control of the researcher
- because of ethical concerns or logistical constraints
- Retrospective studies
 - A retrospective study is performed
 - a posteriori, using information on events that have taken place in the past.
 - *A posteriori*, Latin for "from the latter", is a term from logic, which usually refers to reasoning that works backward from an effect to its causes.
 - in a way, based on reasoning from known facts or past events, rather than by making assumptions or predictions.



2.1 Sample Space

- The set of all possible outcomes of a statistical experiment is called the **Sample Space** and is represented by the symbol S .
- Each outcome in a sample space is called
 - an element
 - or a member of the sample space,
 - or simply a sample point.
- If the sample space has a finite number of elements, we may list the members separated by commas and enclosed in braces.
- Thus, the sample space S , of possible outcomes when a coin is flipped, may be written

$$S = \{H, T\},$$

where H and T correspond to heads and tails, respectively.



2.1 Sample Space

Examples:

- We may record the number of accidents that occur monthly at the intersection of Driftwood Lane and Royal Oak Drive, hoping to justify the installation of a traffic light.
- we might classify items coming off an assembly line as “defective” or “nondefective”.
- we may be interested in the volume of gas released in a chemical reaction when the concentration of an acid is varied.
- We are dealing with either **numerical data**, representing counts or measurements, or **categorical data** both are taken as an **observation**



Definition 2.1:

The set of all possible outcomes of a statistical experiment is called the **sample space** and is represented by the symbol S .

Example 2.1

Consider the experiment of tossing a die.

If we are interested in the number that shows on the top face, the sample space is $S_1 = \{1, 2, 3, 4, 5, 6\}$.

If we are interested only in whether the number is even or odd, the sample space is simply $S_2 = \{\text{even}, \text{odd}\}$.



2.1 Sample Space

- Example 2.1 illustrates the fact that more than one sample space can be used to describe the outcomes of an experiment
- In this case, S_1 provides more information than S_2
- If we know which element in S_1 occurs we can tell which outcome in S_2 occurs
- However, a knowledge of what happens in S_2 is of little help in determining which element in S_1 occurs
- In general, it is desirable to use the sample space that gives the most information concerning the outcomes of the experiment
- In some experiments, it is helpful to list the elements of the sample space systematically by means of a tree diagram



Example 2.2:

An experiment consists of flipping a coin and then flipping it a second time if a head occurs. If a tail occurs on the first flip, then a die is tossed once. To list the elements of the sample space providing the most information, we construct the tree diagram of Figure 2.1.

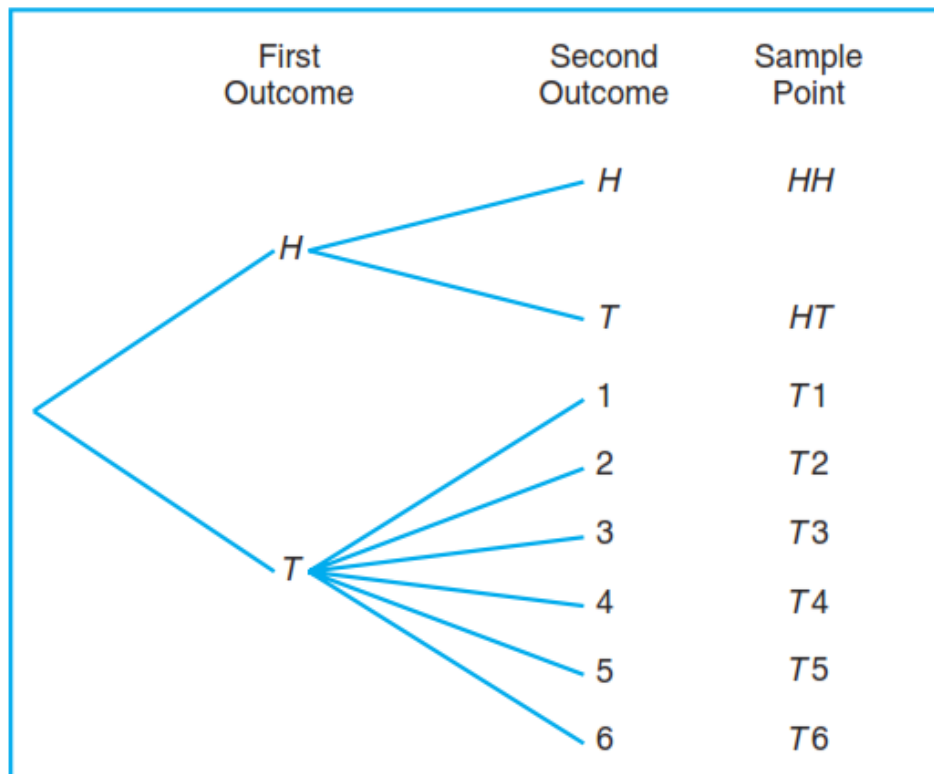


Figure 2.1: Tree diagram for Example 2.2.



The various paths along the branches of the tree give the distinct sample points.

Starting with the top left branch and moving to the right along the first path, we get the sample point HH , indicating the possibility that heads occurs on two successive flips of the coin.

Likewise, the sample point $T3$ indicates the possibility that the coin will show a tail followed by a 3 on the toss of the die.

By proceeding along all paths, we see that the sample space is

$$S = \{HH, HT, T1, T2, T3, T4, T5, T6\}.$$

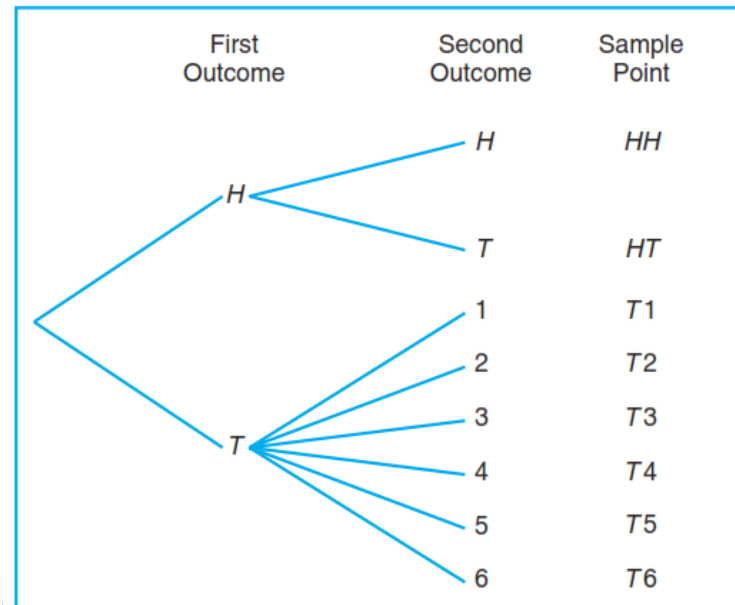


Figure 2.1: Tree diagram for Example 2.2.

Example 2.3:

Suppose that three items are selected at random from a manufacturing process. Each item is inspected and classified defective, D , or nondefective, N . To list the elements of the sample space providing the most information, we construct the tree diagram of Figure 2.2.

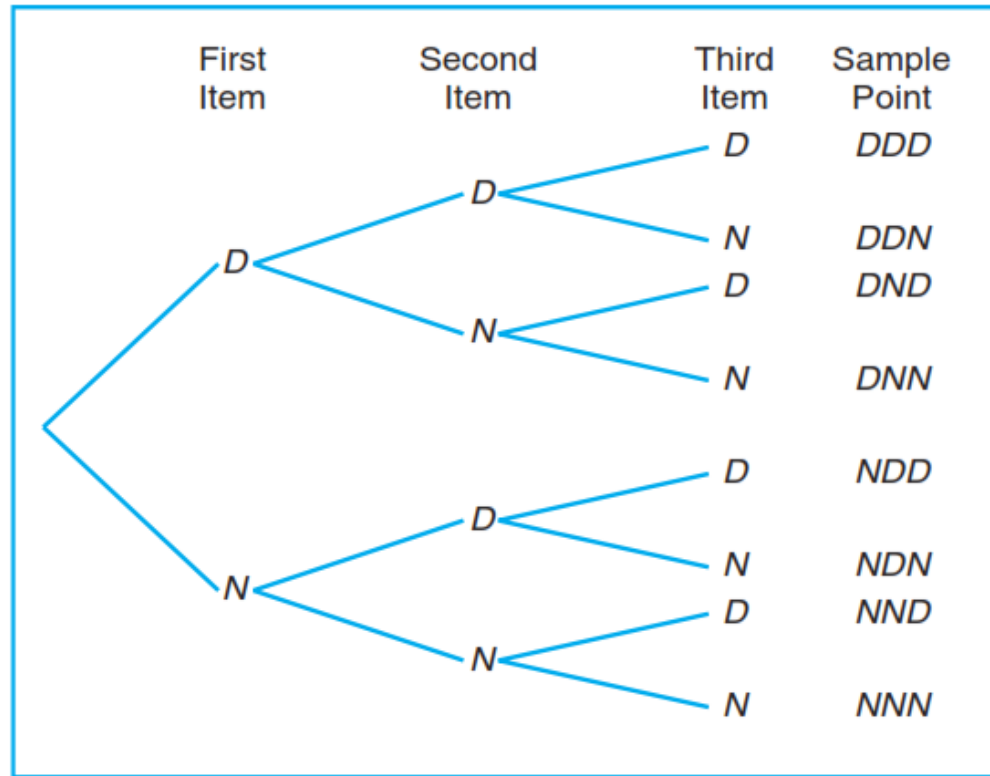


Figure 2.2: Tree diagram for Example 2.3.

Now, the various paths along the branches of the tree give the distinct sample points.



Starting with the first path, we get the sample point DDD , indicating the possibility that all three items inspected are defective.

As we proceed along the other paths, we see that the sample space is

$$S = \{DDD, DDN, DND, DNN, NDD, NDN, NND, NNN\}.$$

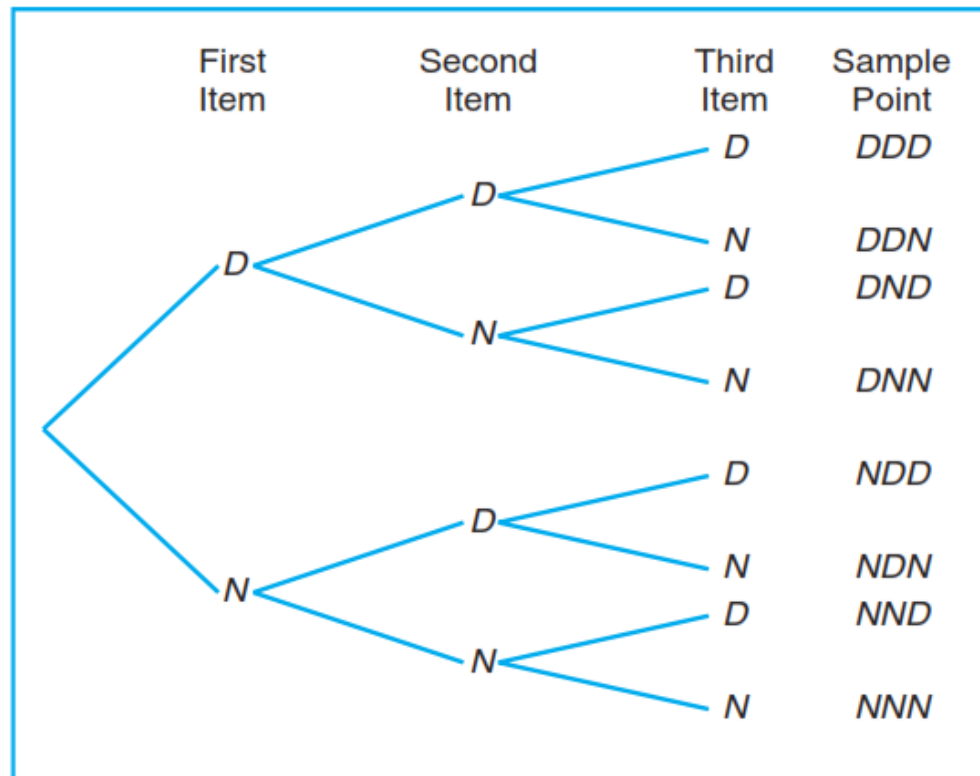


Figure 2.2: Tree diagram for Example 2.3.



Consider the situation of Example 2.3 in which items from a manufacturing process are either D , defective, or N , nondefective. There are many important statistical procedures called sampling plans that determine whether or not a “lot” of items is considered satisfactory. One such plan involves sampling until k defectives are observed. Suppose the experiment is to sample items randomly until one defective item is observed. The sample space for this case is

$$S = \{D, ND, NND, NNND, \dots\}.$$



Sample spaces with a large or infinite number of sample points are best described by a **statement** or **rule method**.

For example, if the possible outcomes of an experiment are the set of cities in the world with a population over 1 million, our sample space is written

$$S = \{x \mid x \text{ is a city with a population over 1 million}\},$$

which reads “ S is the set of all x such that x is a city with a population over 1 million.” The vertical bar is read “such that.”

Similarly, if S is the set of all points (x, y) on the boundary or the interior of a circle of radius 2 with center at the origin, we write the **rule**

$$S = \{(x, y) \mid x^2 + y^2 \leq 4\}.$$

Whether we describe the sample space by the rule method or by listing the elements will depend on the specific problem at hand.

The rule method has practical advantages, particularly for many experiments where listing becomes a tedious chore.



2.2 Event

- For any given experiment, we may be interested in the occurrence of certain events rather than in the occurrence of a specific element in the sample space.
- To each event we assign a collection of sample points, which constitute a subset of the sample space.
- That subset represents all of the elements for which the event is true.
- For instance, we may be interested in the event A that the outcome when a die is tossed is divisible by 3.
- This will occur if the outcome is an element of the subset $A = \{3, 6\}$ of the sample space S in Example 2.1.
- As a further illustration, we may be interested in the event B that the number of defectives is greater than 1 in Example 2.3.
- This will occur if the outcome is an element of the subset
$$B = \{DDN, DND, NDD, DDD\}$$
 of the sample space S



Definition 2.2:

An **event** is a subset of a sample space.

Example 2.4:

Given the sample space $S = \{t \mid t \geq 0\}$, where t is the life in years of a certain electronic component,

then the event A that the component fails before the end of the fifth year is the subset $A = \{t \mid 0 \leq t < 5\}$.



Events

- It is conceivable that an event may be a subset
 - that includes the entire sample space S or
 - a subset of S called the null set and denoted by the symbol ϕ , which contains no elements at all.
- For instance, if we let A be the event of detecting a microscopic organism by the naked eye in a biological experiment, then $A = \phi$.
- Also, if

$$B = \{x \mid x \text{ is an even factor of } 7\},$$

then B must be the null set, since the only possible factors of 7 are the odd numbers 1 and 7.



Events - Complement

- Consider an experiment where the smoking habits of the employees of a manufacturing firm are recorded
- A possible sample space might classify an individual as a non-smoker, a light smoker, a moderate smoker, or a heavy smoker
- Let the subset of smokers be some event
- Then all the non-smokers correspond to a different event, also a subset of S , which is called the **complement** of the set of smokers



Definition 2.3:

The **complement** of an event A with respect to S is the subset of all elements of S that are not in A . We denote the complement of A by the symbol A' .

Example 2.5:

Let R be the event that a red card is selected from an ordinary deck of 52 playing cards, and let S be the entire deck. Then R' is the event that the card selected from the deck is not a red card but a black card.

Example 2.6:

Consider the sample space

$$S = \{\text{book, cell phone, mp3, paper, stationery, laptop}\}.$$

Let $A = \{\text{book, stationery, laptop, paper}\}$. Then the complement of A is $A' = \{\text{cell phone, mp3}\}$.



Intersection

- We now consider certain operations with events that will result in the formation of new events.
- These new events will be subsets of the same sample space as the given events.
- Suppose that A and B are two events associated with an experiment.
- In other words, A and B are subsets of the same sample space S.
- For example, in the tossing of a die we might let A be the event that an even number occurs and B the event that a number greater than 3 shows.
- Then the subsets $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$ are subsets of the same sample space

$$S = \{1, 2, 3, 4, 5, 6\}.$$

- Note that both A and B will occur on a given toss if the outcome is an element of the subset $\{4, 6\}$, which is just the intersection of A and B.



Definition 2.4:

The **intersection** of two events A and B , denoted by the symbol $A \cap B$, is the event containing all elements that are common to A and B .

Example 2.7:

Let E be the event that a person selected at random in a classroom is majoring in engineering, and let F be the event that the person is female. Then $E \cap F$ is the event of all female engineering students in the classroom. ┘

Example 2.8:

Let $V = \{a, e, i, o, u\}$ and $C = \{l, r, s, t\}$; then it follows that $V \cap C = \phi$. That is, V and C have no elements in common and, therefore, cannot both simultaneously occur. ┘



Mutual Exclusivity

- For certain statistical experiments it is by no means unusual to define two events, A and B , that cannot both occur simultaneously.
- The events A and B are then said to be mutually exclusive.
- Stated more formally, we have the following definition:

Definition 2.5:

Two events A and B are **mutually exclusive**, or **disjoint**, if $A \cap B = \phi$, that is, if A and B have no elements in common.



Example 2.9:

A cable television company offers programs on eight different channels, three of which are affiliated with ABC, two with NBC, and one with CBS. The other two are an educational channel and the ESPN sports channel. Suppose that a person subscribing to this service turns on a television set without first selecting the channel. Let A be the event that the program belongs to the NBC network and B the event that it belongs to the CBS network. Since a television program cannot belong to more than one network, the events A and B have no programs in common. Therefore, the intersection $A \cap B$ contains no programs, and consequently the events A and B are mutually exclusive. └



Union

- Often one is interested in the occurrence of at least one of two events associated with an experiment.
- Thus, in the die-tossing experiment, if $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$, we might be interested in either A or B occurring or both A and B occurring.
- Such an event, called the union of A and B , will occur if the outcome is an element of the subset $\{2, 4, 5, 6\}$.

Definition 2.6:

The **union** of the two events A and B , denoted by the symbol $A \cup B$, is the event containing all the elements that belong to A or B or both.



Example 2.10:

Let $A = \{a, b, c\}$ and $B = \{b, c, d, e\}$; then $A \cup B = \{a, b, c, d, e\}$.

Example 2.11:

Let P be the event that an employee selected at random from an oil drilling company smokes cigarettes. Let Q be the event that the employee selected drinks alcoholic beverages. Then the event $P \cup Q$ is the set of all employees who either drink or smoke or do both.

Example 2.12:

If $M = \{x \mid 3 < x < 9\}$ and $N = \{y \mid 5 < y < 12\}$, then

$$M \cup N = \{z \mid 3 < z < 12\}.$$



The relationship between events and the corresponding sample space can be illustrated graphically by means of **Venn diagrams**.

In a Venn diagram we let the sample space be a rectangle and represent events by circles drawn inside the rectangle.

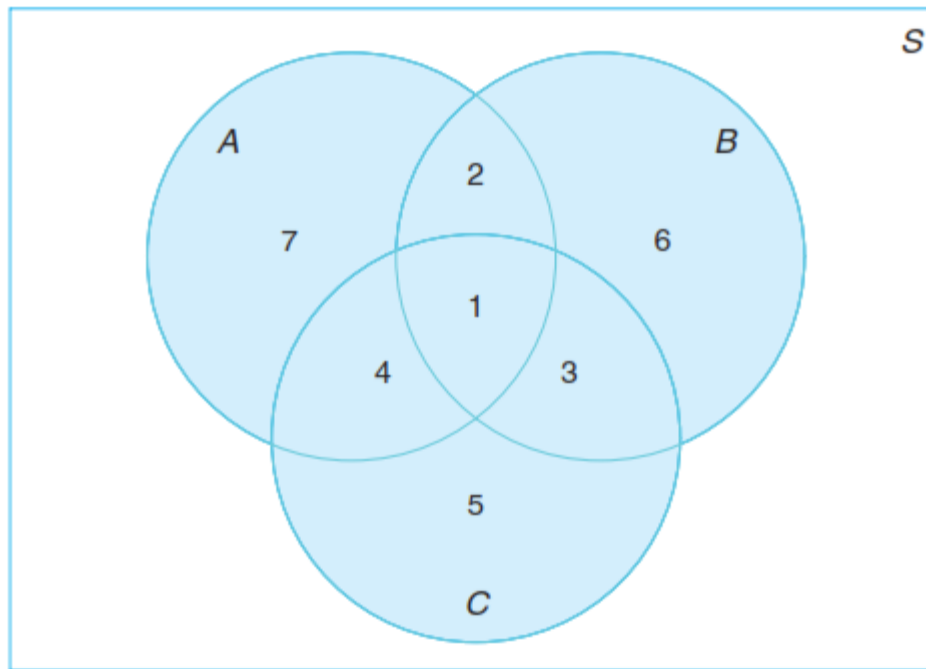


Figure 2.3: Events represented by various regions.

Thus, in Figure 2.3, we see that $A \cap B =$ regions 1 and 2, $B \cap C =$ regions 1 and 3,



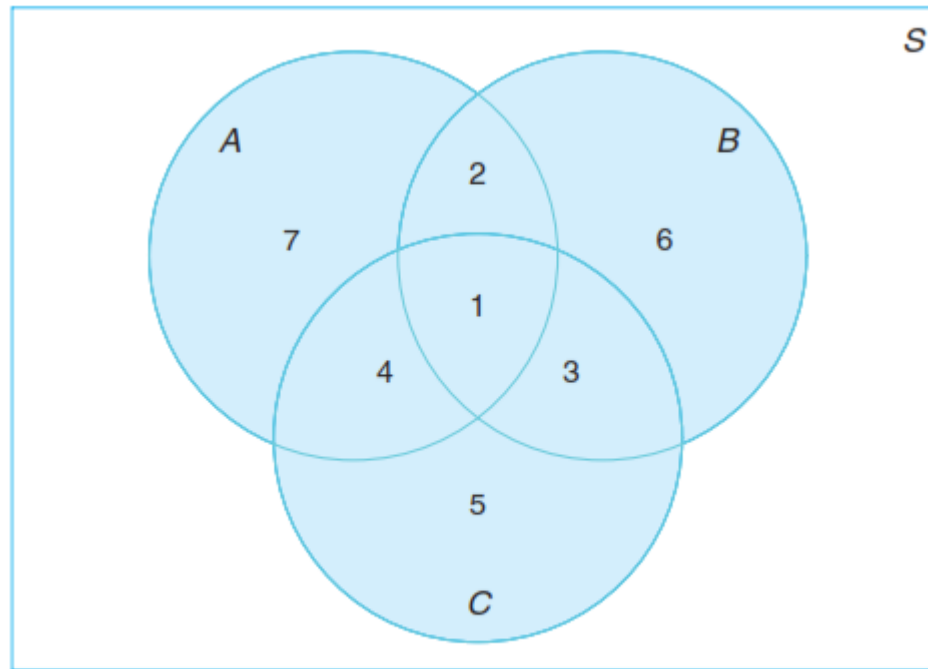


Figure 2.3: Events represented by various regions.

$$B' \cap A = \text{regions 4 and 7,}$$

$$A \cap B \cap C = \text{region 1,}$$

$$(A \cup B) \cap C' = \text{regions 2, 6, and 7,}$$

$$A \cup C = \text{regions 1, 2, 3, 4, 5, and 7,}$$



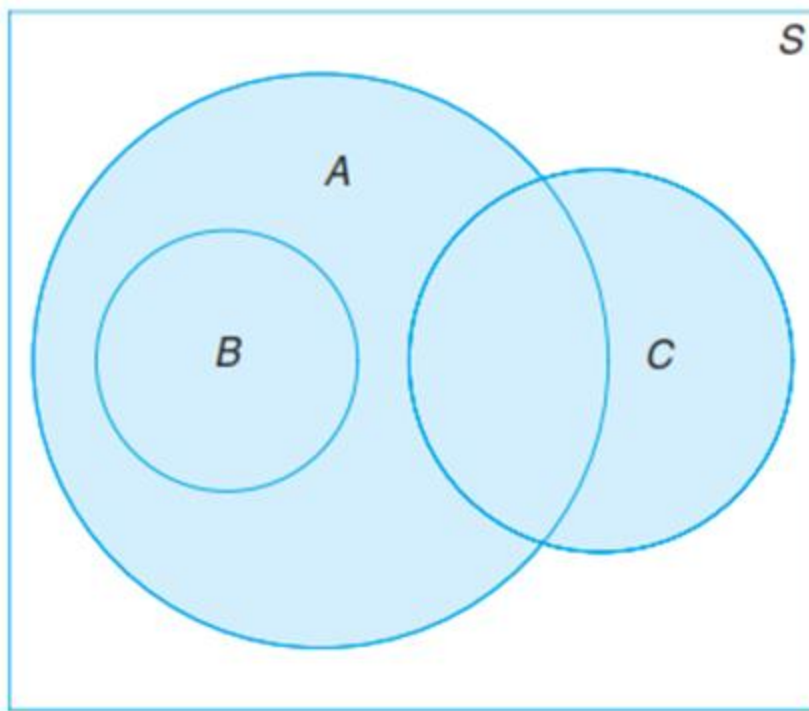


Figure 2.4:
Events of the sample space S .

In Figure 2.4, we see that events A , B , and C are all subsets of the sample space S .

It is also clear that event B is a subset of event A ;

event $B \cap C$ has no elements and hence B and C are mutually exclusive;

event $A \cap C$ has at least one element;

and event $A \cup B = A$.

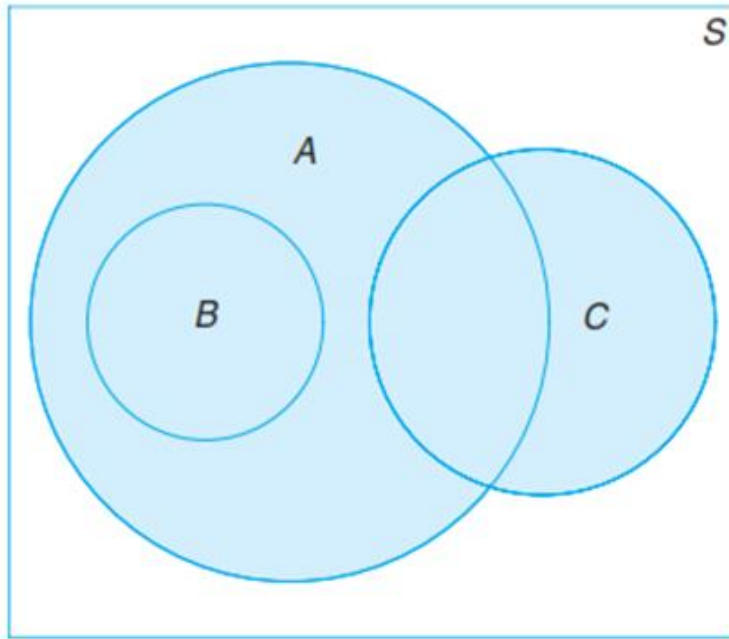


Figure 2.4:
Events of the sample space S .

Figure 2.4 might, therefore, depict a situation where we select a card at random from an ordinary deck of 52 playing cards and observe whether the following events occur:

A : the card is red,

B : the card is the jack, queen, or king of diamonds,

C : the card is an ace.

Clearly, the event $A \cap C$ consists of only the two red aces.

Several results that follow from the foregoing definitions, which may easily be verified by means of Venn diagrams, are as follows:

1. $A \cap \phi = \phi$.

2. $A \cup \phi = A$.

3. $A \cap A' = \phi$.

4. $A \cup A' = S$.

5. $S' = \phi$.

6. $\phi' = S$.

7. $(A')' = A$.

8. $(A \cap B)' = A' \cup B'$.

9. $(A \cup B)' = A' \cap B'$.



Question 1

List the elements of each of the following sample spaces:

- (a) the set of integers between 1 and 50 divisible by 8;
- (b) the set $S = \{x \mid x^2 + 4x - 5 = 0\}$;
- (c) the set of outcomes when a coin is tossed until a tail or three heads appear;
- (d) the set $S = \{x \mid x \text{ is a continent}\}$;
- (e) the set $S = \{x \mid 2x - 4 \geq 0 \text{ and } x < 1\}$.



Question 2:

An experiment involves tossing a pair of dice, one green and one red, and recording the numbers that come up. If x equals the outcome on the green die and y the outcome on the red die, describe the sample space S

- (a) by listing the elements (x, y) ;
- (b) by using the rule method.



Question 2:

- For the sample space
- (a) list the elements corresponding to the event A that the sum is greater than 8;
 - (b) list the elements corresponding to the event B that a 2 occurs on either die;
 - (c) list the elements corresponding to the event C that a number greater than 4 comes up on the green die;
 - (d) list the elements corresponding to the event $A \cap C$;
 - (e) list the elements corresponding to the event $A \cap B$;
 - (f) list the elements corresponding to the event $B \cap C$;
 - (g) construct a Venn diagram to illustrate the intersections and unions of the events A , B , and C .

